EXPERIMENT-08
Collection of Email Evidence from a Windows PC

Aim

To capture a Windows computer's RAM using DumpIt and analyze the resulting memory image with Bulk Extractor Viewer to identify email-related artifacts.

Tools Required:

1. DumpIt
2. Bulk Extractor and Bulk Extractor Viewer

Procedure :

Step 1: Capture RAM Using DumpIt

- Obtain authorization to examine the Windows computer.
- Prepare DumpIt and an external destination drive with sufficient free space.
- Run DumpIt with the required privileges.
- Confirm the memory acquisition prompt by entering Y if requested.
- Wait for the acquisition to complete.
- Record the output RAW filename, file size, acquisition time, and any reported errors.

Step 2: Locate the Memory Dump

- Open File Explorer.
- Navigate to the directory where DumpIt saved the output.
- Confirm that the RAW file exists and record its size.
- Secure the dump and calculate a SHA-256 hash where possible.

Step 3: Open Bulk Extractor Viewer

- Obtain and install or extract a compatible version of Bulk Extractor and its viewer.
- Launch the appropriate application.
- Select the acquired RAW memory image as the input.

- Choose a separate output directory for the analysis results.

Step 4: Analyze the Memory Image

- Start the extraction process using the available options.
- Wait for processing to finish.
- Review the generated feature files and extraction logs.
- Locate email.txt if the email-address feature was generated.
- Open email_histogram.txt if available to review email-related frequency statistics.

Step 5: Record the Findings

- Record the number of email-related artifacts found.
- Review the extracted results and distinguish valid addresses from false positives.
- Preserve the analysis output and document the tool version and processing status.
- Include screenshots of the acquisition, analysis, and results in the lab record.

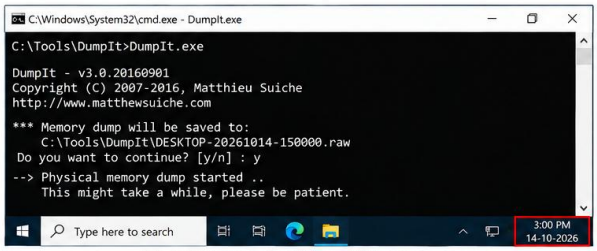


Figure 1: Running DumpIt for RAM Capture

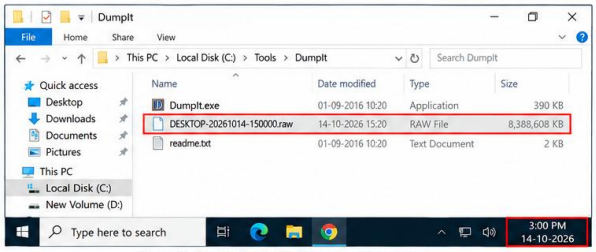


Figure 2: Generated .RAW Memory Dump File

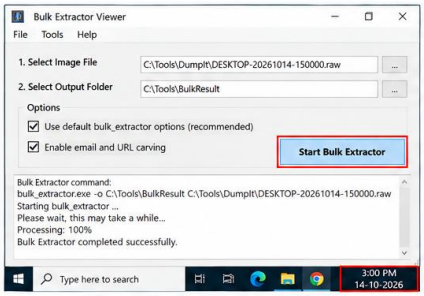


Figure 3: Running Bulk Extractor Viewer

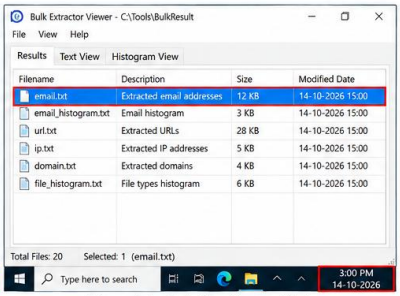


Figure 4: Analysis Results – email.txt

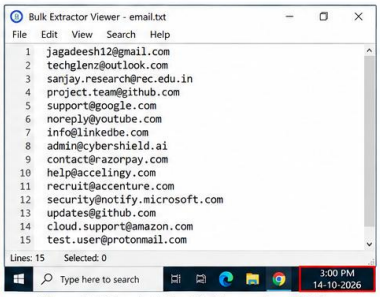


Figure 5: Extracted Email Addresses (email.txt)

Result

The procedure for capturing physical memory using DumpIt and examining a memory image with Bulk Extractor Viewer was studied. Email-related artifacts were identified as a potential source of evidence.